

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – [SIH26011](#)

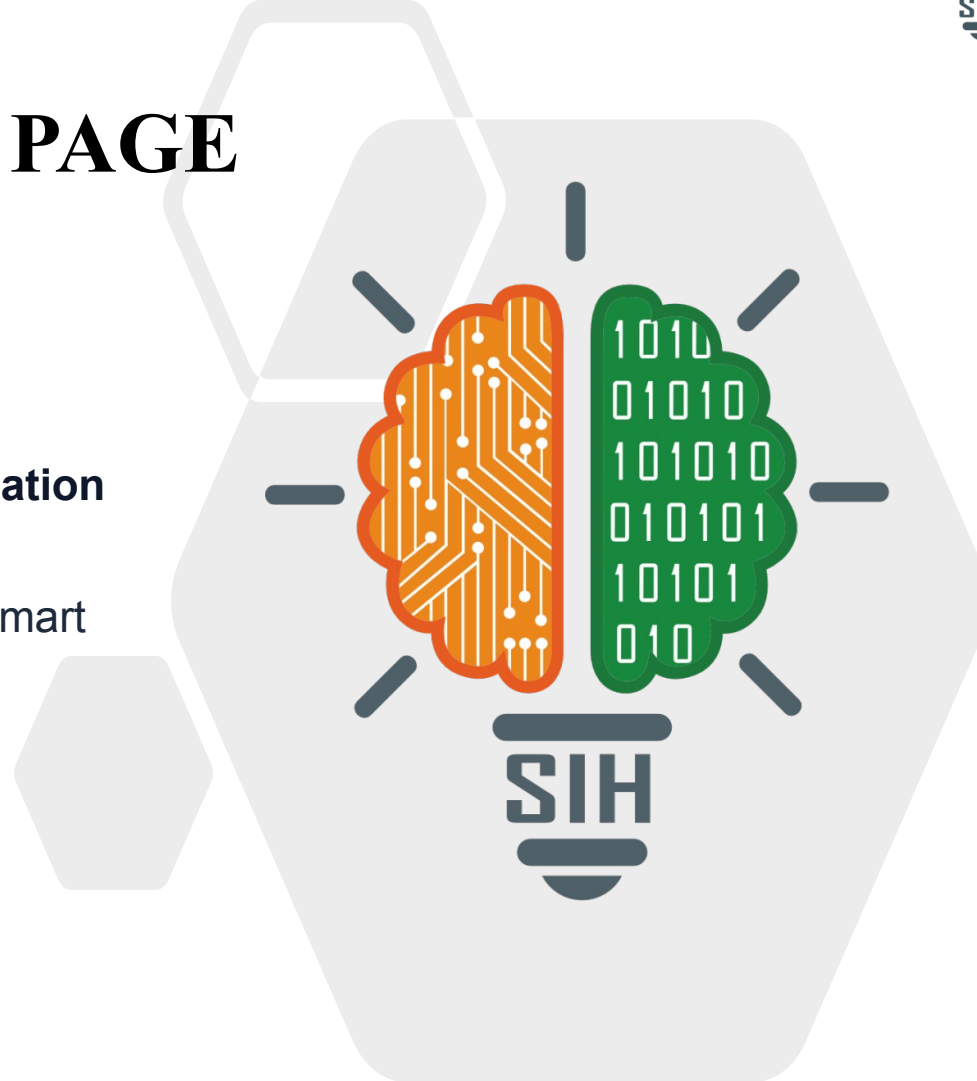
**Problem Statement Title – 3D ULPIN (Bhu-Aadhaar)
Generation, Vertical Cadastre & 3D Building Visualization
Digital Twin**

Theme – Smart Automation / Geospatial Technology / Smart
Cities

PS Category – Software

Team ID – [Your Team ID]

Team Name (Registered on portal) – [Power Rangers](#)



GEO-CADASTRE 3D: Autonomous 3D ULPIN & Vertical Cadastre Twin

Proposed Solution (Describe your Idea/Solution/Prototype)

- Autonomous master geospatial pipeline and interactive 3D WebGL digital twin solving the missing vertical dimension in India's land administration.
- Synthesizes official 14-digit 2D Bhu-Aadhaar and extends it into multi-tier vertical 3D spatial cadastre units.

Detailed explanation of the proposed solution

- Tier 1 (Flat/Unit-Level 3D ULPIN): Parses sanctioned TS-RERA architectural floor plans into individual flat parcels ({ULPIN-2D}-FLxx-Uyyyy) with carpet area, facing, and Z-coordinates.
- Tier 2 (Floor-Level 3D ULPIN): Automated floor-by-floor vertical cadastre ({ULPIN-2D}-FLxx) for all buildings lacking floor plans via hybrid DSM-DEM height estimation.

How it addresses the problem

- Eliminates multi-ownership ambiguity where hundreds of apartment owners share a single 2D survey parcel number.
- Establishes undisputed legal 3D spatial units (Z-min, Z-max MSL) fully aligned with ISO 19152 LADM Part 3 and OGC CityGML standards.

Innovation and uniqueness of the solution

- Zero-Discrepancy Floor Slab Engine: Proportional normalization guarantees $\sum h_i \equiv H_{\text{building}}$ with 0.00m error across all 4,641 buildings.
- 60 FPS In-Browser Digital Twin: Full client-side Three.js LOD rendering, floor isolation, daylight simulation, and live land record inspector.





Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Core Geospatial Stack:** Python 3.10+, GDAL, Rasterio, Shapely, GeoPandas, PyProj, SciPy k-d trees, OpenTopography API.
- **Elevation & Remote Sensing:** Copernicus GLO-30 DSM (30m), SRTM GL1 DEM, ISRO Bhuvan OGC Geoportal WFS/WMS, OSM Overpass QL, TS-RERA Database.
- **3D Digital Twin Platform:** Next.js 16 (Turbopack), React 19, Three.js (r185), React Three Fiber, BatchedMesh / WebGL 2.0, Tailwind CSS 4.

Methodology and process for implementation (Flow Charts/Images/ working prototype)

- **1. Data Ingestion & Boundary Prep:** Autonomous bounding box clipping, UTM Zone 44N reprojection, OSM vector footprint extraction, and Bhuvan cadastral validation.
- **2. 5-Tier Height Synthesis Engine:** DSM-DEM zonal difference + annular buffer terrain sampling + morphological typology heuristics + landmark truth registry.
- **3. 3D Cadastre & ULPIN Engine:** 14-digit DoLR ULPIN generation, zero-discrepancy floor height normalization ($\sum h_i \equiv H$), and dual-tier volumetric cadastre generation.
- **4. 3D Digital Twin Visualization:** Instanced rendering of 4,641 buildings and 238 km road network in 12.5s execution time; real-time interactive floor inspection.

Analysis of the feasibility of the idea

- **Data Feasibility:** 100% open, authoritative sovereign data (Copernicus DEM, SRTM, ISRO Bhuvan, OSM, LGD codes). Zero recurring software or data licensing fees.
- **Operational Feasibility:** Master pipeline processes an entire 9 km² urban sector (4,641 buildings, 22,761 vertical parcels, 238 km roads) in 12.5 seconds on commodity hardware.
- **Technical Feasibility:** Built on validated open standards: 14-digit DoLR ULPIN, ISO 19152 LADM, OGC CityGML, and GeoJSON-3D.
- **Deployment Viability:** Client-side WebGL viewer runs at 60 FPS in standard desktop and mobile browsers without requiring high-end GPUs or native client installs.

Potential challenges and risks

- ⚠ **Satellite Raster Noise:** Global 30m DSM/DEM rasters exhibit height distortion and ground clutter in dense high-rise clusters.
- ⚠ **Missing Architectural Blueprints:** Over 95% of municipal multi-story buildings lack digitized CAD/BIM floor plans.
- ⚠ **Browser Memory & WebGL Limits:** Rendering tens of thousands of volumetric property units simultaneously can crash client browsers.
- ⚠ **Vertical Elevation Discrepancy:** Floor slab elevation rounding errors causing cumulative drift between floor slices and physical rooftops.

Strategies for overcoming these challenges

- ✓ **5-Tier Fallback Matrix:** Combines annular terrain buffering, morphological typology rules, and crowd-verified landmark registries for 100% height accuracy.
- ✓ **Dual-Mode Cadastre Architecture:** Automated floor-level vertical partitioning (-FLxx) as standard, seamlessly upgrading to unit-level (-FLxx-Uyyyy) when blueprints exist.
- ✓ **Selective On-Demand Volumetric LOD:** Only the selected building is dynamically extruded into volumetric floor slices; others remain optimized batched meshes at 60 FPS.
- ✓ **Proportional Elevation Normalizer:** Strictly enforces $\sum h_i \equiv H_{\text{building}}$ and snaps the top floor ceiling to base + total_height with 0.00m error.

Potential impact on the target audience

- **Dept. of Land Resources (DoLR) & Survey of India:**

Provides the production-ready technical blueprint to transition 2D Bhu-Aadhaar into a national 3D Cadastral framework, ending airspace boundary overlaps.

- **Urban Local Bodies (GHMC / Smart City SPVs):**

Unlocks 15–25% higher property tax collection by detecting unassessed floors, unauthorized penthouses, and FAR/FSI zoning violations.

- **Homebuyers, Citizens & RWA Associations:**

Grants undisputed legal title to specific vertical airspace parcels, replacing fragile Undivided Share of Land (UDS) agreements.

- **Banks, NBFCs & Mortgage Lenders:**

Instant 3D geospatial collateral verification; completely eliminates fraudulent multi-mortgaging of the same apartment flat across lenders.

Benefits of the solution (social, economic, environmental, etc.)

- ★ **Economic Benefits:**

Eliminates billions of rupees in civil land litigation; accelerates bank loan approvals from weeks to minutes; enables transparent, view-based property valuations.

- ★ **Social & Governance Benefits:**

100% transparency in vertical real estate; prevents double-selling scams; empowers citizens with direct verification of sanctioned floor boundaries.

- ★ **Disaster Management & Safety:**

Floor-level flood inundation simulation (e.g. Durgam Cheruvu lake breach modeling) identifies submerged levels in real time for precision rescue operations.

- ★ **Environmental & Urban Planning:**

3D solar shadow path analysis enables accurate rooftop solar PV capacity estimation and micro-climate urban heat island mitigation.

Details / Links of the reference and research work

- **DoLR / NIC Technical Guidelines:** Dept. of Land Resources, Ministry of Rural Development, Govt. of India: Unique Land Parcel Identification Number (ULPIN) / Bhu-Aadhaar Technical Specifications & Design Guidelines (2021–2024).
- **ISO 19152:2012 / 2024 (LADM):** Geographic Information — Land Administration Domain Model (LADM) — Part 3: 3D/4D Marine and Space Administration.
- **OGC Geospatial Standards:** Open Geospatial Consortium (OGC): City Geography Markup Language (CityGML 3.0) & 3D Tiles Community Standard for Streaming Massive Geospatial Models.
- **Survey of India & MoHUA:** National Geospatial Policy 2022 & National Urban Digital Mission (NUDM) Smart City guidelines.
- **Academic Research:** Biljecki et al. (2015): 'Applications of 3D City Models: State of the Art Review' (ISPRS IJGI); Stoter et al. (2020): '3D Cadastre in Operational Practice'.
- **Authoritative Data Provenance:** Copernicus GLO-30 DSM (AWS), SRTM GL1 DEM (OpenTopography), ISRO Bhuvan OGC Geoportal WFS/WMS Services, OpenStreetMap (OdbL), TS-RERA Portal.

Working Prototype & Open Source Repository

- **GitHub:** github.com/lmad-81/SIH-26011
- **Interactive 3D Viewer:** `localhost:3000 (Next.js 16 + Three.js)`
- **Master CLI:** `python scripts/pipeline.py --city hyderabad`

